FIVUCSAS

Face and Identity Verification Using Cloud-based SaaS

The numbers behind a multi-tenant biometric auth platform

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0%
Replay-attack false-accept
n = 120 recorded-video attempts

<1s
End-to-end verification
P95 = 950 ms, CPU only

10
Composable auth factors
any subset → MFA flow per tenant

At a glance

Attacks defeated

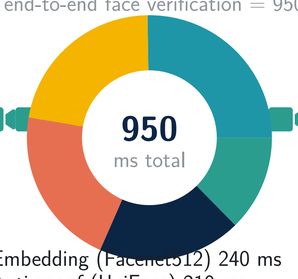
each icon = a confirmed defeat in our test bench

- Photo on screen 30 / 30
- Recorded video 60 / 60
- Printed photo 15 / 15
- Static 3D mask 13 / 15
- Deepfake stream ongoing

Defended by sequence-mismatch heuristic; under continued red-team

Latency budget

P95 end-to-end face verification = 950 ms



- Embedding (Facenet512) 240 ms
- Anti-spoof (UniFace) 210 ms
- DB search (HNSW) 200 ms
- Face mesh (468 pt) 180 ms
- Face detection (MTCNN) 120 ms

The codebase

by the numbers, 2026-05

1 820+ tests

633 Java + 619 web + 425 mobile + 114 spoof-detector + 27 Playwright

5 services **60 migrations**

8 ADRs **1 CX43 VPS**

Hetzner Cloud · 8 CPU · 16 GB RAM · 150 GB · Ubuntu 24.04

Docker 29.3.0 · Traefik 3.6 · Postgres 17 + pgvector · Redis 7.4

Contributions

C1. Tenant-configurable composition of **10 auth factors** into MFA flows declared at admin layer.

C2. Active randomised **liveness** — n-step facial-action sequence bound to a per-attempt nonce.

C3. Pipeline decomposition along the **OIDC / ML iteration** boundary, two services on private network.

C4. Embedding-at-rest **Fernet encryption** (AES-128-CBC + HMAC-SHA-256) prior to pgvector write.

C5. Regulator-aware data lifecycle — **GDPR Art. 15 / 17** and KVKK soft-delete + purge + audit.

M. Per-attempt method — server issues nonce, client streams 60 frames, server checks EAR/MAR/yaw + spoof.

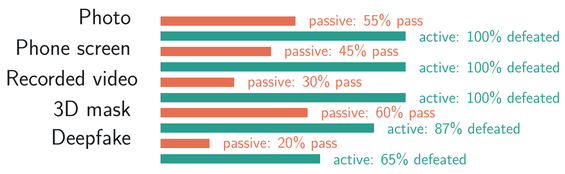
10 composable factors

Password · Email OTP · SMS OTP · TOTP · QR code · Face · Voice · Fingerprint · WebAuthn · NFC document

gold = active-liveness wrapped

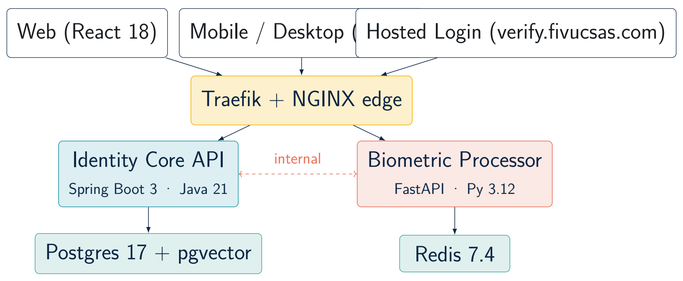
Why active liveness

Spoof type vs. defence layer



Read: passive bars = share that *still passes* when only passive checks are used. Active bars = share *defeated* when the randomised challenge is in front. Higher active is better.

System map



Biometric service has no public route — Docker network only, API-key gated.

Conclusion

Production-grade, GDPR-compliant biometric authentication is achievable on commodity hardware when the system is decomposed along its right axis — a frozen OIDC contract on one side, an iterating ML pipeline on the other — and when liveness is treated as an active, challenge-driven property rather than a passive signal alone. Measured ***0 / 120 replay-attack false-accepts*** *and* ***0.97 passive-spoof AUC*** *are both within practical deployment limits on a single CPU server.*

Live demo & what's next

Scan to test it
`verify.fivucsas.com`

Future work

- → BYOD tenant DB — tenants host biometric data themselves
- → On-device pre-filter — bandwidth + privacy win
- → Cross-modal challenge — voice nonce + facial action
- → W3C verifiable credentials post-KYC

References

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[2] Lugaresi, C. et al. *MediaPipe.* arXiv:1906.08172.
[3] Soukupová, T., Čech, J. *Real-Time Eye Blink Detection.* CVWW, 2016.
[4] Wan, L. et al. *Generalized End-to-End Loss for Speaker Verification.* ICASSP, 2018.
[5] Yadav, S. et al. *UniFace.* CVPR-W, 2023.

github.com/Rollingcat-Software/FIVUCSAS · fivucsas.com · ahmetabdullahgultekin@gmail.com · Marmara University 2026